

Unveiling care patterns: Unsupervised machine learning clustering of epilepsy hospitalizations in the Brazilian public health system (2021-2025)

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BACKGROUND: Epilepsy represents a significant burden on the Brazilian Unified Health System (SUS), characterized by regional disparities in access and quality of care. Traditional epidemiological analyses often rely on geographic division (Macro-regions), which may mask heterogeneities in hospital efficiency and outcomes. Unsupervised Machine Learning techniques, such as clustering, offer a data-driven approach to identify latent patterns of care, grouping states by performance similarity rather than geography. **OBJECTIVE:** To identify and characterize distinct profiles of hospital assistance for epilepsy in Brazil using the K-Means clustering algorithm, based on efficiency, cost, and mortality indicators. **METHODS:** This is a cross-sectional ecological study using data from the Hospital Information System (SIH/SUS). We analyzed all hospitalizations due to Epilepsy (ICD-10 G40-G41) across 27 Brazilian federative units from 2021 to 2025. Three Key Performance Indicators (KPIs) were calculated for each state: Hospital Mortality Rate (%), Mean Cost per Admission (R\$), and Mean Length of Stay (days). Data were normalized using Z-score standardization. The K-Means algorithm was applied to group states into clusters. The optimal number of clusters (K=4) was determined using the Elbow Method and validated by the Silhouette Score. **RESULTS:** A total of 320,419 epilepsy-related hospitalizations were analyzed. The K-Means analysis (Silhouette Score = 0.316) identified four distinct assistance profiles. Cluster 0 "Efficient Care" (n=12; e.g., Minas Gerais, Rio Grande do Sul, Amazonas) demonstrated the best performance, with the lowest mortality rate (2.11%) and shortest length of stay (5.1 days). Cluster 1 "High Resource" (n=8; e.g., São Paulo, Bahia, Goiás) presented significantly higher mean costs (R\$1,279) and mortality (3.55%). Cluster 2 "Critical" (n=1, Alagoas) figures more than double the national average mortality rate (6.93%) with the highest mean cost (R\$2,349). Finally, Cluster 3 "Low Cost/Moderate Outcome" (n=6; e.g., Rio de Janeiro, Pernambuco) was characterized by the lowest mean cost (R\$ 713) but intermediate mortality rates (3.46%). **CONCLUSION:** The application of Machine Learning identified deep disparities in epilepsy care that transcend geographical regions. The algorithm distinguished between high-efficiency systems and critical outliers requiring immediate audit. These findings demonstrate the utility of AI in health management, suggesting that public policies should target specific cluster profiles rather than broad regional strategies to optimize resource allocation and reduce mortality.

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Anexos importantes:

